Proof skeleton

Exact Diffusion Design for Maximally Collective Stable Turing Patterns
in Binary-Complex Mass-Action Networks
Topology-Wide All-Spectrum Localization and Exponent-Optimal Stationary Heterogeneity Trade-Offs

1. Reaction algebra and topology-wide localization

Construct Y_m and Γ_m directly from the indexed reactions. Reaction-by-reaction verification gives $c^T\Gamma_m = 0$ for $c = (0, 4, \ldots, 4, 2, 1)^T$. Solving the balance equations gives exactly

$$\ker \Gamma_m = \operatorname{span}\{(\mathbf{1}_m, 0, 0), (0_m, 1, 1)\}.$$

Because Γ_m has $m + 2$ columns, rank–nullity gives $\operatorname{rank} \Gamma_m = m$. Independently, the maximal minor on rows $X_1, X_3, \ldots, X_m, Z$ and columns $R_2, \ldots, R_{m-2}, R_a, R_b, R_+$ has determinant $4(-1)^m$. Therefore every positive flux is $(a\mathbf{1}_m, b, b)$. Differentiation at the unit equilibrium gives $A_m(a, b)$; at a general equilibrium $x^* = H^{-1}\mathbf{1}$, the Jacobian is $A_m(a, b)H$. Conversely, the choice

$$k_r = \frac{v_r}{(x^*)^{y_r}}$$

realizes every $a, b, H > 0$.

For a nonempty principal set with fewer than m vertices, Frobenius decomposition reduces every SCC to either a negative singleton or one of three non-singleton forms: the $X_1, \ldots, X_{m-1}$ long cycle, the $X_2, \ldots, X_m$ long cycle, or a principal boundary-triad block. At $b = 2a$, the deleted edge $X_1 \to X_m$ lies on neither long cycle and can only refine the boundary decomposition. The two cycle characteristic equations are excluded from $\Re\lambda \geq 0$ by strict product inequalities. The full triad has positive cubic coefficients and $c_1c_2 - c_3$ is a sum of fourteen positive monomials. Every lower-order principal matrix is thus Hurwitz. For the core determinant, the interior lower-bidiagonal Schur block has determinant $(-a)^{m-3}$ and bottom-left inverse entry $-1/a$; the remaining 3×3 matrix has determinant $2a^2b$. Restoring the column factors gives the negative signed determinant $-2a^{m-1}b\prod h_i$, forcing a positive real eigenvalue.

For $m = 3$, the two long cycles are $\{X_1, X_2\}$ and $\{X_2, X_3\}$, and every other non-singleton SCC is a principal subgraph of $\{X_1, X_3, Z\}$. Thus the chain argument is applied only for $m \geq 4$.

Audit object. Independently generate every induced SCC for $m = 3, \ldots, 8$ at generic b and at $b = 2a$, while the proof uses the all-m graph classification rather than enumeration.

2. Principal-minor diffusion-ray theorem and exact network law

Let $n \geq 2$. For $a_I = (-1)^{|I|} \det J_{I,I}$ and $D \succ 0$, multilinearity gives

$$\det(sD - J) = \sum_I a_I \prod_{j \notin I} sd_j = s(\beta_1 + \beta_2 s + \cdots + \beta_n s^{n-1}).$$

Assume directly that $a_I > 0$ for $|I| \leq n-2$ and that $\sum_{|I|=n-1} a_I > 0$; a simple zero and otherwise stable spectrum supplies the latter coefficient condition in the application. Hence $\beta_k > 0$ for $k \geq 2$, and the bracket is strictly increasing on $s > 0$, proving the unique simple threshold iff $\beta_1 < 0$.

For the stronger positive-real band, define

$$\chi_s(\lambda) = \det(\lambda I + sD - J) = \sum_I a_I \prod_{j \notin I} (\lambda + sd_j).$$

Every derivative contribution from $|I| \leq n-2$ is nonnegative. The total order-$(n-1)$ derivative is $\sum_{|I|=n-1} a_I > 0$, the coefficient of λ in $\det(\lambda I - J)$. Hence $\chi_s'(\lambda) > 0$ for $\lambda \geq 0$. The sign of $\chi_s(0)$ therefore gives a positive real eigenvalue exactly for $0 < s < s_*$. At threshold, $\chi_{s_*}(0) = 0$ and $\chi_{s_*}'(0) > 0$; this, rather than scalar simplicity of the generalized-pencil root alone, proves ordinary algebraic simplicity of zero for $J - s_* D$.

For $A_m(a,b)H$, compute every order-m omission minor. Omitting Z gives weight -2; omitting an interior X_j gives $+16$; omitting X_1 or X_m gives zero. Thus

$$\beta_1 = 2a^{m-1} b \prod_{i=1}^{m} h_i \left(8h_Z \sum_{j=2}^{m-1} \frac{d_j}{h_j} - d_Z \right).$$

Homogeneous stability on $c^\perp$, together with $c^T J = 0$, gives rank $n-1$, an algebraically simple conservation zero, and a positive order-$(n-1)$ characteristic coefficient, so every hypothesis of the general theorem holds. This proves the exact criterion. The criterion is independent of a, b, but its unique root is $s_*(a,b,H,D)$. The same sign follows from first-order perturbation of the conservation eigenvalue. For fixed H, minimizing $d_{\max}/d_{\min}$ under the strict inequality gives the nonattained infimum $T(H) = 8h_Z \sum_{j=2}^{m-1} h_j^{-1}$. At $H = I$, $T = 8(m-2)$, and $h_Z/h_j \geq 1/\chi_H$ gives $\chi_D \chi_H > 8(m-2)$.

3. Improved unit-equilibrium primary bifurcation

Set $a = b = 1$, $H = I$, $K_i = 91m - 181 - i$, and use

$$d_1 = 23/63, \quad d_i = K_i^{-1}, \quad d_m = 1/7, \quad d_Z = 16/45.$$

The full vectors $r = (r_1, \dots, r_m, r_Z)^T$ and $\ell = (\ell_1, \dots, \ell_m, \ell_Z)^T$ satisfy $(A_m - D_m)r = 0$ and $(A_m - D_m)^T\ell = 0$. Exact harmonic-sum formulas prove $\ell^T r < 0$ and $\ell^T D_m r < 0$, hence transversality $\eta_m > 0$.

Homogeneous stability follows from

$$\det(\lambda I - A_m) = \lambda q_m(\lambda), \qquad \lambda q_m = (1+\lambda)^{m-3}P(\lambda) - R(\lambda),$$

where $P = \lambda^4 + 12\lambda^3 + 42\lambda^2 + 47\lambda + 16$ and $R = 5\lambda^2 + 33\lambda + 16$. The exact difference $|1+\lambda|^2|P|^2 - |R|^2$ expands into 35 nonnegative monomials in $x = \Re\lambda$ and $y^2 = (\Im\lambda)^2$, with equality only at the origin. For $m = 3$, $q_3 = (\lambda+7)(\lambda^2+5\lambda+2)$ is Hurwitz. For $m \geq 4$, $|1+\lambda| \geq 1$ makes the 35-term strict inequality incompatible with $(1+\lambda)^{m-3}P = R$ at any nonzero closed-right-half-plane root. Since $q_m(0) = 16m - 34 > 0$, the conservation zero is simple.

For spatial damping tD_m, set $g_1 = \lambda+2+(23/63)t$, $g_m = \lambda+5+t/7$, $g_Z = \lambda+4+(16/45)t$, $F = g_1 g_m g_Z - 4g_1 - 4g_m + g_Z$, and $G = g_Z(4g_1 + g_m) - 36$. Chain elimination gives $\det(\lambda I - A_m + tD_m) = Q_m F - G$. At $(\lambda, t) = (0, 1)$, the interior product telescopes to $91/90$. With $z = y^2$ and $s = t - 1$, the exact source

$$E_{77} = \left[(91/90)^2 + z\right] |F(x + i\sqrt{z}, 1+s)|^2 - |G(x + i\sqrt{z}, 1+s)|^2$$

has 77 strictly positive coefficients and no constant term. Thus equality occurs only at $(x, z, s) = (0, 0, 0)$. Algebraic simplicity at that point is the separate exact identity, with $\mathfrak{h}_m = \sum_{j=1}^{m-2} K_j^{-1}$,

$$\frac{d}{d\lambda}\det(\lambda I - A_m + D_m)\bigg|_{\lambda=0} = \frac{7043400m - 13600927 - 7043400\mathfrak{h}_m}{255150} = -\frac{163}{45}\ell^T r > 0.$$

Hence the first mode has one algebraically simple zero with one-dimensional kernel and cokernel, all other first-mode eigenvalues are stable, and all $t > 1$ modes are Hurwitz.

The exact unit contrast is

$$\chi_D = \frac{23(91m - 183)}{63},$$

which improves the superseded high-contrast profile and is within a dimension-independent factor of the sharp linear threshold.

4. Conservation-compatible normal form and stable branch

The mass-action field is quadratic:

$$f(\mathbf{1}+u)=A_m u+\tfrac12 B_m(u,u),\qquad c^T B_m(u,v)=0.$$

On the fixed integrated-mass class, the critical mode is $r\cos\xi$. Since $\cos^2\xi=(1+\cos 2\xi)/2$, solve

$$A_m w_0=-\tfrac14 B(r,r),\quad c^T w_0=0,\qquad (A_m-4D_m)w_2=-\tfrac14 B(r,r).$$

The first is an augmented nonsingular solve; the second is invertible by mode isolation. The interior w_2 recurrence telescopes through four consecutive K_i factors, leaving an explicit 4×4 system in the four retained boundary values. Its determinant is $64\mathcal{Q}_m/[6615(91m-183)(91m-181)(91m-180)]>0$.

The fixed-mass sectorial semiflow has a one-dimensional center manifold. The spatial reflection $\xi\mapsto\pi-\xi$ commutes with the Neumann semiflow, preserves the mass class, and sends $r\cos\xi$ to its negative. An equivariant center manifold therefore has an odd reduced vector field, and reflection exchanges the two nonzero branches. Their perturbations tend to zero in $H^2_N\hookrightarrow C^0$ from $\mathbf{1}$, so both branches remain componentwise strictly positive for sufficiently small $\mu>0$ at each fixed m. Projection of its reduced vector field onto the adjoint critical mode yields

$$\dot A=\eta_m\mu A+c_m A^3+O(|A|^5+|A|^3|\mu|+|A||\mu|^2),$$

$$c_m=\frac{\ell^T\{B(r,w_0)+\frac12 B(r,w_2)\}}{\ell^T r}.$$

Setting this reduced vector field to zero gives the stationary Lyapunov–Schmidt equation. Writing $\mathfrak{h}_m=\sum_{j=1}^{m-2}K_j^{-1}$, the numerator telescopes to $R_m+C_m\mathfrak{h}_m$. The supplement prints R_m, C_m, their positive denominators, and the exact L_m clearing identity. Shifted coefficientwise-positive polynomials prove the numerator positive, while $\ell^T r<0$; therefore $c_m<0$ for every $m\ge 3$. For a direct audit, put

$$\mathcal{Q}_m=589180301m^3-3500015940m^2+6930529579m-4574434500,$$

$$\begin{aligned}P_R&=68605040480814208768m^4-550882186169626030957m^3+1658612632937449670852m^2\\&\quad-2219226476204103501323m+1113379274975809565700,\\P_C&=652054120726848m^4-5151971981328467m^3+15265080924982572m^2\\&\quad-20102347725659113m+9927281930180400.\end{aligned}$$

Then

$$R_m=\frac{P_R}{286118780220(8m-17)\mathcal{Q}_m},\qquad C_m=-\frac{215P_C}{11645046(8m-17)\mathcal{Q}_m},$$

and exact clearing gives

$$R_m+C_m\frac{m-2}{90m-179}=\frac{L_m}{286118780220(8m-17)(90m-179)\mathcal{Q}_m}.$$

The shifted coefficients of $\mathcal{Q}_m, P_R, P_C, L_m$ are positive, so $C_m<0$ and $R_m+C_m\mathfrak{h}_m\ge R_m+C_m(m-2)/(90m-179)>0$.

On

$$X_c=\{u\in H^2_N:\textstyle\int_0^\pi c^T u=0\},\qquad Y_c=\{g\in L^2:\int_0^\pi c^T g=0\},$$

fixed-mass invariance follows from $c^T A_m=0$ and the Neumann boundary conditions. The onset linearization $\mathcal{L}_0=A_m+D_m\partial_{\xi\xi}$ has cosine blocks $A_m-k^2D_m$: $k=0$ is invertible on $c^\perp$, $k=1$ has its certified simple zero, and every $k\ge 2$ block is invertible. For large k,

$$A_m-k^2D_m=-k^2D_m(I-k^{-2}D_m^{-1}A_m),\qquad \|(A_m-k^2D_m)^{-1}\|\le 2\|D_m^{-1}\|k^{-2}.$$

Thus compatible L^2 data have an H^2_N preimage, the range is closed, and the stationary linearization is Fredholm of index zero, with kernel span$\{r\cos\xi\}$, cokernel span$\{\ell\cos\xi\}$, and transversality pairing $(\pi/2)\ell^T D_m r\ne 0$. Crandall–Rabinowitz gives the stationary local curve; reflection equivariance and the odd normal form identify its two positive nonconstant halves for $\mu>0$. Their center eigenvalue is $-2\eta_m\mu+O(\mu^2)<0$. The bound $\alpha(A_m-k^2D_m)\le\|(A_m+A_m^T)/2\|_2-k^2d_{\min}\to-\infty$ reduces the complementary onset gap to finitely many modes. Positive Neumann diffusion is sectorial with compact resolvent on fixed-mass L^2; along the branch its positive diffusion lower bound controls high frequencies, while analytic perturbation preserves the finitely many low spectral gaps. Its fractional H^1 domain is a Banach algebra in one dimension, and the quadratic Nemytskii map is smooth. The principle of linearized stability gives local exponential convergence in H^1 for sufficiently close nonnegative initial data in the same conservation class.

5. Equilibrium-scaled stable trade-off and audit boundaries

Let $\nu = m - 2$ and

$$\kappa_\nu = \begin{cases} 1/\sqrt{3}, & \nu = 1, \\ \sqrt{5}/2, & \nu \geq 2, \end{cases} \qquad L_0 = \frac{\kappa_\nu}{\sqrt{\nu}}, \qquad L_1 = \frac{90\nu}{90\nu+1}.$$

The interval is nonempty: for $\nu = 1$ the squared inequality is $8281 < 24300$, while for $\nu \geq 2$ it follows from $(90\nu+1)^2 \leq (91\nu)^2 < 6480\nu^3$. Set $h_1 = h_m = h_Z = 1$, $h_i = K_i/(LK_{i-1})$, and choose physical diffusion $D^{\mathrm{phys}} = \mathsf{H}_m(L)\Delta$. Under $\widehat{x} = \mathsf{H}_m(L)x$,

$$\partial_t \widehat{x} = \mathsf{H}_m(L)\{f(\widehat{x}) + (1-\mu)\Delta\partial_{xx}\widehat{x}\}.$$

At $\mu = 0$, the effective stationary bracket and right vector are unchanged; the dynamic left vector is $\widetilde{\ell}(L) = \mathsf{H}_m(L)^{-1}\ell$. The physical fixed-mass functional is $(\mathsf{H}_m(L)^{-1}c)^T\widehat{x}$. Moreover,

$$\widetilde{\ell}(L)^T\mathsf{H}_m(L)\Delta r = \ell^T\Delta r < 0, \qquad \widetilde{\ell}(L)^T r < 0,$$

so the scaled crossing coefficient $\eta_m(L)$ is positive. Invertibility of $\mathsf{H}_m(L)$ preserves the kernel:

$$\ker[\mathsf{H}_m(L)(A_m - \Delta)] = \ker(A_m - \Delta) = \operatorname{span}\{r\}.$$

A generalized vector would force $0 = \widetilde{\ell}(L)^T r$, so the scaled zero is algebraically simple. For each fixed m and L, the normalized branch converges to $\mathbf{1}$ in $H^2_N \hookrightarrow C^0$ and remains componentwise positive for small $\mu > 0$; the positive diagonal physical rescaling preserves this positivity.

Chain elimination now gives interior factors

$$g_i = \frac{K_{i-1}}{K_i}(1 + L\lambda) + \frac{t-1}{K_i}.$$

For $t \geq 1$, Bernoulli's inequality and $\nu L^2 \geq 1/3$ reduce spatial half-plane exclusion to the exact source polynomial

$$E_{84} = (91/90)^2(1 + Ax + z/3)|F(x+i\sqrt{z}, 1+s)|^2 - |G(x+i\sqrt{z}, 1+s)|^2, \quad A = 2\nu L, z = y^2, s = t-1.$$

Its 84 coefficients are coefficientwise nonnegative in $A > 0$, and equality occurs only at the first-mode zero $(x, z, s) = (0, 0, 0)$.

At $t = 0$, the exact homogeneous determinant, after dividing by $\det \mathsf{H}_m(L)$, is

$$QF_0 - R, \qquad Q = \prod_{i=2}^{m-1}\left(1 + L\frac{K_{i-1}}{K_i}\lambda\right),$$

$$F_0 = \lambda^3 + 11\lambda^2 + 31\lambda + 16, \qquad R = 5\lambda^2 + 33\lambda + 16.$$

Every factor of Q has modulus at least $|1 + L\lambda|$ for $\Re\lambda = x \geq 0$. If $\nu \geq 2$, then

$$|Q|^2 \geq 1 + 2\nu Lx + \nu L^2y^2 \geq 1 + Ax + \frac{5}{4}y^2, \qquad A = 2\nu L.$$

On the certified domain $A \geq \sqrt{5\nu} \geq \sqrt{10} > 1/4$. With $z = y^2$ and $U = A - 1/4$, the exact source

$$E_{22} = [1 + (U + 1/4)x + (5/4)z]\,|F_0(x+i\sqrt{z})|^2 - |R(x+i\sqrt{z})|^2$$

has 22 nonzero monomials. All coefficient polynomials are nonnegative and equality is possible only at $\lambda = 0$. The coefficient $5/4$ is sharp for this certificate only: replacing it by B makes the $z = y^2$ coefficient at $x = 0$ equal to $256B - 320$. This does not identify an intrinsic dynamical boundary. Also

$$(QF_0 - R)'(0) = 16L\sum_{i=2}^{m-1} K_{i-1}/K_i - 2 > 0,$$

so the conservation zero is simple. If $\nu = 1$, putting $\gamma = 91L/90$ gives

$$\frac{QF_0 - R}{\lambda} = \gamma\lambda^3 + (1 + 11\gamma)\lambda^2 + (6 + 31\gamma)\lambda + (16\gamma - 2),$$

whose coefficients and Routh–Hurwitz difference $6 + 99\gamma + 325\gamma^2$ are positive because $\gamma > 1/8$.

The second harmonic is unchanged, while the zero harmonic changes by $w_0(L) = w_0^{\mathrm{ref}} + \tau_m(L)\rho$. The cubic numerator is $N_m(L) = N_m^{\mathrm{ref}} + \tau_m(L)S_m$. With $1/91 \le \mathfrak{h}_m < 1/90$,

$$S_m = -\frac{4(1760850\mathfrak{h}_m - 10253)}{462105}, \qquad -\frac{1}{10} < S_m < 0.$$

Since $C_m < 0$, exact clearing gives

$$N_m^{\mathrm{ref}} - \frac{1}{100} \ge \frac{P_{\mathrm{ref}}(\nu)}{D_{\mathrm{ref}}(\nu)},$$

where

$$\begin{aligned}
P_{\mathrm{ref}} &= 37905029866372656848 40\nu^5 - 97421653046860028648 9\nu^4 - 53103567440921218871\nu^3 \\
&\quad - 576386186827093561\nu^2 + 3649732858601219\nu + 55281268032918, \\
D_{\mathrm{ref}} &= 715296950550(8\nu - 1)(90\nu + 1) \\
&\quad \times (589180301\nu^3 + 35065866\nu^2 + 629431\nu + 3306).
\end{aligned}$$

$D_{\mathrm{ref}} > 0$, and the shift $\nu = v + 1$ gives six positive coefficients for P_{ref}, proving $N_m^{\mathrm{ref}} > 1/100$.

For the gauge, set

$$\begin{aligned}
A_\tau &= 1494249120\mathfrak{h}_m L\nu^2 - 69786990\mathfrak{h}_m L\nu + 108738630L\nu^2 \\
&\quad + 1214388L\nu - 8521L - 125249670\nu^2 + 1031940\nu, \\
B_\tau &= 32760\mathfrak{h}_m L\nu + 32760L\nu^2 + 4L - 4095\nu.
\end{aligned}$$

Then $\tau_m = -A_\tau/[15876(8\nu - 1)B_\tau]$ and $B_\tau = 4095\nu[8L(\nu + \mathfrak{h}_m) - 1] + 4L > 0$. The exact derivatives are negative on $L \ge 1/\sqrt{3\nu}$, so the maximum is at $(\mathfrak{h}_m, L) = (1/91, 1/\sqrt{3\nu})$. For $\nu \ge 2$ its endpoint sign is bounded by

$$P_{\mathrm{up}}(\nu) = -\frac{189709065}{2}\nu^3 - 507201030\nu^2 - 935658\nu + 58481,$$

with

$$P_{\mathrm{up}}' = -\frac{569127195}{2}\nu^2 - 1014402060\nu - 935658 < 0, \quad P_{\mathrm{up}}(2) = -2789453215 < 0.$$

The case $\nu = 1$ uses the exact unreduced endpoint expression and $\sqrt{3} < 7/4$. Hence $\tau_m(L) < 1/20$. These bounds hold on the broader cubic range $L \ge 1/\sqrt{3\nu}$ and give $N_m(L) > 1/200$. Since $\widetilde{\ell}(L)^T r = \ell^T \mathsf{H}_m(L)^{-1} r < 0$, the exact quotient

$$c_m(L) = \frac{N_m(L)}{\widetilde{\ell}(L)^T r} < 0$$

holds throughout the certified interval.

The physical contrasts are

$$\chi_D = \frac{23}{63}91\nu L, \qquad \chi_H = \frac{91\nu - 1}{91\nu L}.$$

Here χ_H measures equilibrium concentration-scale separation, not a universal measure of biological expense. Moreover

$$\chi_D(L)\chi_H(L) = \frac{23(91\nu - 1)}{63} = \frac{23(91m - 183)}{63} = \chi_D^{\mathrm{unit}}(m).$$

Thus the family reallocates a fixed product without reducing it. Throughout the certified interval, $\chi_D(L) > \chi_H(L)$, so $\max(\chi_D, \chi_H) = \chi_D(L)$ is uniquely minimized at $L = L_0$. At L_0,

$$\chi_D = \frac{2093}{63}\kappa_\nu\sqrt{\nu}, \qquad \chi_H = \frac{\sqrt{\nu}}{\kappa_\nu}\frac{91\nu - 1}{91\nu},$$

so both, and hence their maximum, are $\Theta(\sqrt{m})$, compared with $\Theta(m)$ for the unit-equilibrium stable profile. The topology-specific necessary bound for stationary crossings implies any such realization must have $\max(\chi_D, \chi_H) > \sqrt{8\nu}$, so exponent $1/2$ is optimal among stationary-crossing realizations of $\widehat{N}_m$. Constants remain open.

Failure tests. Verify the SCC exhaustion at $b = 2a$; the principal-minor derivative argument; the 22/35/77/84-term equality cases and the direct $\nu = 1$ cubic; the transformed physical diffusion; the fixed-mass gauge; and the comparison $N_m(L) > 1/200$. No finite regression substitutes for these all-m identities.